

TuneWorks

Technical Report

Group 1:

Vivienne Gunadhi, Kailyn Tai, Xiangying Qi, Claire Yang, Vera Mao, Sylvia Hou, Jinghong Chen,
Kangni Chen

Table Overview:

Agents: Contains core information about sales agents and their employment details.

Customers: Stores customer information including identity, location, and contact details.

Engagements: Records every entertainment engagement, linking customers, agents, entertainers, and contract details.

Entertainers: Holds basic details for each entertainer, including identification, location, and contact information.

Entertainer_Styles: Records the musical styles associated with each entertainer, along with the strength or proficiency of each style.

Entertainer_Members: Lists the members associated with each entertainer group, including member IDs and their status.

Members: Contains basic demographic information about individuals who may belong to entertainer groups.

Musical_Preferences: Tracks customers' preferred musical styles and the order of their preferences.

Musical_Styles: Definitions of musical styles available.

ztbldays: Date dimension table for daily analysis.

ztblmonths: Month-level date dimension table.

Ztblweeks: Week-level date dimension table.

ztblSkipLabels: Helper table for labels.

Data Dictionary:

1. Agents

Column	Type	Definition / Business Meaning
agentid	INT	Unique identifier for each sales agent.
agtfirstname	TEXT	Agent's first name.
agtlastname	TEXT	Agent's last name.
agtstreetaddress	TEXT	Agent's street address.
agtcity	TEXT	Agent's city of residence.
agtstate	TEXT	Agent's state of residence.
agtzipcode	TEXT/INT	Agent's ZIP code.

agtphonenumber	TEXT	Agent's phone number.
datehired	DATE	Date the agent was hired.
salary	INT	Base salary of the agent.
commissionrate	INT	Commission rate applied to agent sales.

2. Customers

Column	Type	Definition / Business Meaning
customerid	INT	Unique identifier for each customer.
custfirstname	TEXT	Customer first name.
custlastname	TEXT	Customer last name.
custstreetaddress	TEXT	Customer street address.
custcity	TEXT	Customer city.
custstate	TEXT	Customer state.
custzipcode	TEXT/INT	Customer ZIP code.
custphonenumber	TEXT	Customer phone contact.

3. Engagements:

Column	Type	Definition / Business Meaning
engagementnumber	INT	Unique ID for each entertainment engagement (booking).
startdate	DATE	Date the engagement begins.
enddate	DATE	Date the engagement ends.
starttime	TIME	Start time of the engagement.
stoptime	TIME	End time of the engagement.
contractprice	INT	Total contract price charged to the customer.
customerid	INT	Linked customer who booked the engagement.
agentid	INT	Linked sales agent who arranged the engagement.
entertainerid	INT	Entertainer hired for the engagement.

4. Entertainers

Column	Type	Definition / Business Meaning
entertainerid	INT	Unique identifier for each entertainer.
entstagename	TEXT	Stage name of the entertainer.
entssn	TEXT	Social Security Number (for payroll).
entstreetaddress	TEXT	Address of the entertainer.
entcity	TEXT	City of the entertainer.
entstate	TEXT	State of the entertainer.
entzipcode	TEXT/INT	ZIP code.
entphonenumber	TEXT	Phone number.
entwebpage	TEXT	Entertainer's website.
entemailaddress	TEXT	Entertainer's email.
dateentered	DATE	Date entertainer entered the system.

5. Entertainer_Members

Column	Type	Definition / Business Meaning
entertainerid	INT	Linked entertainer/group that the member belongs to.
memberid	INT	Linked member identifier.
status	INT	Indicates active/inactive or role status within the group (1/2 etc.).

6. Members

Column	Type	Definition / Business Meaning
memberid	INT	Unique member identifier.
mbrfirstname	TEXT	Member's first name.
mbrlastname	TEXT	Member's last name.
mbrphonenumber	TEXT	Member phone number.
gender	TEXT	Gender identification (M/F/null).

7. Musical_Preferences

Column	Type	Definition / Business Meaning
customerid	INT	Customer expressing the musical preference.
styleid	INT	Musical style selected.
preferenceseq	INT	Rank of preference (1 = top choice).

8. Musical_Styles

Column	Type	Definition / Business Meaning
styleid	INT	Unique musical style ID.
stylename	TEXT	Name of the musical style (e.g., jazz, classical).

9. Entertainer_Styles

Column	Type	Definition / Business Meaning
entertainerid	INT	Entertainer who is associated with the musical style.
styleid	INT	Musical style performed.
stylestrength	INT	Rating/strength for how skilled the entertainer is at this style.

10. ztblDays

Column	Type	Definition / Business Meaning
datefield	DATE	Daily date dimension for analysis.

11. ztblMonths

Column	Type	Definition / Business Meaning
monthyear	TEXT	Month name/year label.
yearnumber	INT	Year (e.g., 2017).
monthnumber	INT	Month integer (1-12).
monthstart	DATE	First day of the month.
monthend	DATE	Last day of the month.
january	INT	Month is January.

february	INT	Month is February.
march	INT	Month is March.
april	INT	Month is April.
may	INT	Month is May.
june	INT	Month is June.
july	INT	Month is July.
august	INT	Month is August.
september	INT	Month is September.
october	INT	Month is October.
november	INT	Month is November.
december	INT	Month is December.

12. ztblWeeks

Column	Type	Definition / Business Meaning
weekstart	DATE	Start date of a calendar week.
weekend	DATE	End date of a calendar week.

13. ztblSkipLabels

Column	Type	Definition / Business Meaning
labelcount	INT	Helper field used for labeling/formatting.

Ambiguous Fields:

1. Entertainer_Members.Status

- The Status field in the Entertainer_Members table contains numeric codes (1, 2) with no documentation indicating their meaning
- It is difficult to be certain of the meaning of status.

2. Agents / Entertainers vs Engagements vs ztblDays/Months

- Agents.DateHired (1997–1998)/ Entertainers.DateEntered (1995–1998)
- Engagements.StartDate Engagements.EndDate (2017–2018) / ztblDays.DateField (2017-01-01 - 2020-01-01) / ztblMonths (2017-01 - 2019-12)
- Potential Concerns:
 - Are there missing earlier engagements?
 - Are the DatesHired/DatesEntered columns labeled incorrectly?

Potential Input Issues/Missing Data:

1. Input Issues

a. AGENTS AgentID = 9

- i. AgtFirstName = 'Daffy', AgtLastName = 'Dumbwit'
- ii. Salary = 50
- iii. CommissionRate = 0.01
- iv. Salary and commission rate are less than other agents. Is this a test input that was not removed because their lastname is Dumbwit and address is 1234 Main Street.

b. ENTERTAINERS EntertainerID = 1011

- i. EntCity = 'Seattle', EntState = 'WA'
- ii. EntZipCode = '99837' (Comparing to 980xx or 981xx)
- iii. The Zipcode is for Alaska and not Seattle, Washington.

c. ENGAGEMENTS Engagementnumber=23

- i. Startdate='2017-10-01', Starttime=20:00:00
- ii. Enddate='2017-10-01', Endtime=0:00:00
- iii. The enddate is before the start date.

d. ENGAGEMENTS Engagementnumber=90

- i. Startdate='2018-01-09', Starttime=20:00:00
- ii. Enddate='2018-01-09', Endtime=2:00:00
- iii. The enddate is before the start date.

e. ENGAGEMENTS Engagementnumber=118

- i. Startdate='2018-02-19', Starttime=19:00:00
- ii. Enddate='2018-02-19', Endtime=0:00:00
- iii. The enddate is before the start date.

2. Missing Data

a. MEMBERS MemberID = 125

- i. Gender = NULL

b. ENTERTAINERS EntertainerID = 1002

- i. EntEmailAddress = NULL

c. ENTERTAINERS EntertainerID = 1004

- i. EntWebPage = NULL
- ii. EntEmailAddress = NULL

d. ENTERTAINERS EntertainerID = 1005

- i. EntEmailAddress = NULL

e. ENTERTAINERS EntertainerID = 1007

- i. EntEmailAddress = NULL

f. ENTERTAINERS EntertainerID = 1008

- i. EntWebPage = NULL
- ii. EntEmailAddress = NULL

g. ENTERTAINERS EntertainerID = 1009

- i. EntWebPage = NULL

h. ENTERTAINERS EntertainerID = 1011

- i. EntWebPage = NULL

- ii. EntEmailAddress = NULL
 - i. **ENTERTAINERS EntertainerID = 1013**
 - i. EntWebPage = NULL
- 3. Inconsistencies**
- a. **Streetaddress for AGENTS/CUSTOMERS/ENTERTAINERS inconsistent format**
 - i. AGENTS: 30301 - 166th Ave. N.E. vs 16679 NE 41st Court
 - ii. CUSTOMERS: 4110 Old Redmond Rd. vs 2222 Springer Road
 - iii. ENTERTAINERS: Route 2, Box 203B
 - b. **Jim Glynn (Entertainer) and Liz Keyser (Customer) inconsistent zipcode**
 - i. STREETADDRESS: 13920 S.E. 40th Street
 - ii. CITY: Bellevue
 - iii. JIM GLYNN ZIPCODE: 98009
 - iv. LIZ KEYSER ZIPCODE: 98006
- 4. Unknown Issue**
- a. **EngagementNumber not complete (Missing 1, 18, 20)**
 - i. EngagementNumber is non-contiguous. It could be either due to missing data or misinputted data.

Resolving Data Quality Issues

- 1. Streetaddress Inconsistencies:**
 - a. Cannot fix without external source, not essential for analysis
- 2. Check Duplicates**
 - a. **Customers:**

```
select custfirstname, custlastname, count(*) as cnt
from customers
group by custfirstname, custlastname
having count(*) > 1
```
 - b. **Entertainers:**

```
select entstagename, count(*) as cnt
from entertainers
group by entstagename
having count(*) > 1
```
 - c. **Members:**

```
select mbrfirstname, mbrlastname, count(*) as cnt
from members
group by mbrfirstname, mbrlastname
having count(*) > 1
```
- 3. Why is the date hired /date entered between 1997–2000 but engagements only in 2017–2018?**
 - a. **Possible explanation #1:** Agent/Entertainer hire dates are historical records. Engagements may only cover the most recent two years of business activity.
 - b. As the date hired/date entered was not used in the analysis, left as is.

4. Ambiguous status in Entertainer_member

- a. Assumption:
 - i. 1: Active member in entertainer group
 - ii. 2: Non-active member in entertainer group

5. Missing webpages and email addresses for entertainers

- a. May be expected for certain groups.
- b. Not required for analysis.

6. Missing gender

- a. Updated through SQL query:
update members
set gender = 'U'
where gender is null
-- where U = Unknown

7. Agents AgentID = 9

- a. As the commission rate and the salary are outliers and the name, address, and phone number seem like a potential test entry, AgentID=9's data was removed from analysis.
- b. SQL Query:
delete from agents where agentid=9

8. Renumber Engagements

- a. Given consecutive numbering based off the original order to simplify analysis
- b. SQL Query:
delete from engagements where engagementnumber in (23,90,118)

```
with ordered AS (select engagementnumber, row_number() over
(order by engagementnumber) as newnum
from engagements)
update engagements e
set engagementnumber = o.newnum
from ordered o
where e.engagementnumber = o.engagementnumber
```

Additional Data To Improve Analysis:

1. From original data - apply for more consistency

- a. Missing data for entertainment number
- b. Emails and the website

2. Additional data

- a. Entertainers' staying duration
 - i. For the risk of the remaining members leaving (Stability)
- b. Customers demographics (Age / Income / Budget Range)
 - i. For promotion in the future
- c. Entertainers' demographics
 - i. Understand more of the members' personal information to be able to target our audience

- d. Entertainers' popularity ranking from customers and members
 - i. Might want to incorporate similar entertainers to join

Analysis and Visualizations:

Analysis was separated into sections based on focus data.

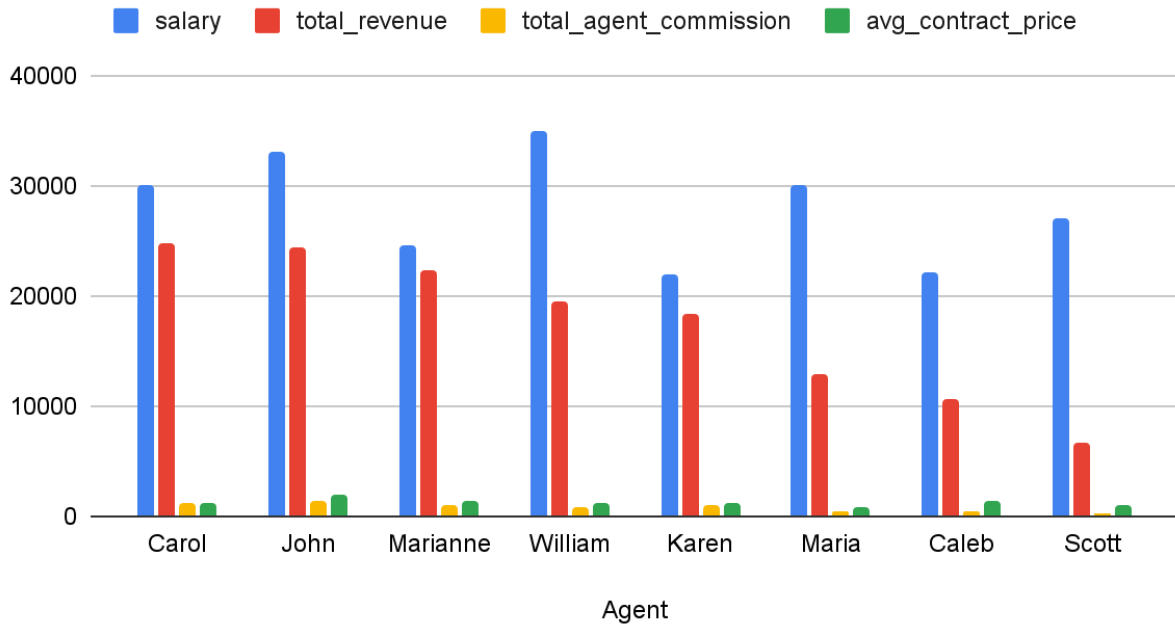
1. Agents

a. Top performing agents

```
select a.agtfirstname, a.agtlastname, a.salary, sum(e.ContractPrice) as Total_Revenue,
commissionrate, Round(sum(e.ContractPrice)*a.commissionrate) as
Total_Agent_Commission, count(*) as Num_Of_Engagements,
round(avg(e.ContractPrice)) as Avg_Contract_Price from Engagements as e
join Agents as a on e.Agentid = a.Agentid
group by a.AgtFirstName, a.AgtLastName, a.salary, a.commissionrate
order by Total_Revenue desc
```

agtfirstna me	agtlastname	salary	total_r evenue	commiss ionrate	total_agent_ commission	num_of_en gagements	avg_contr act_price
Carol	Viescas	30000	24800	0.05	1240	19	1305
John	Kennedy	33000	24435	0.06	1466	12	2036
Marianne	Wier	24500	22315	0.045	1004	17	1313
William	Thompson	35000	19545	0.04	782	15	1303
Karen	Smith	22000	18305	0.055	1007	16	1144
Maria	Patterson	30000	12825	0.04	513	15	855
Caleb	Viescas	22100	10645	0.035	373	8	1331
Scott	Bishop	27000	6720	0.04	269	6	1120

Top Performing Agent

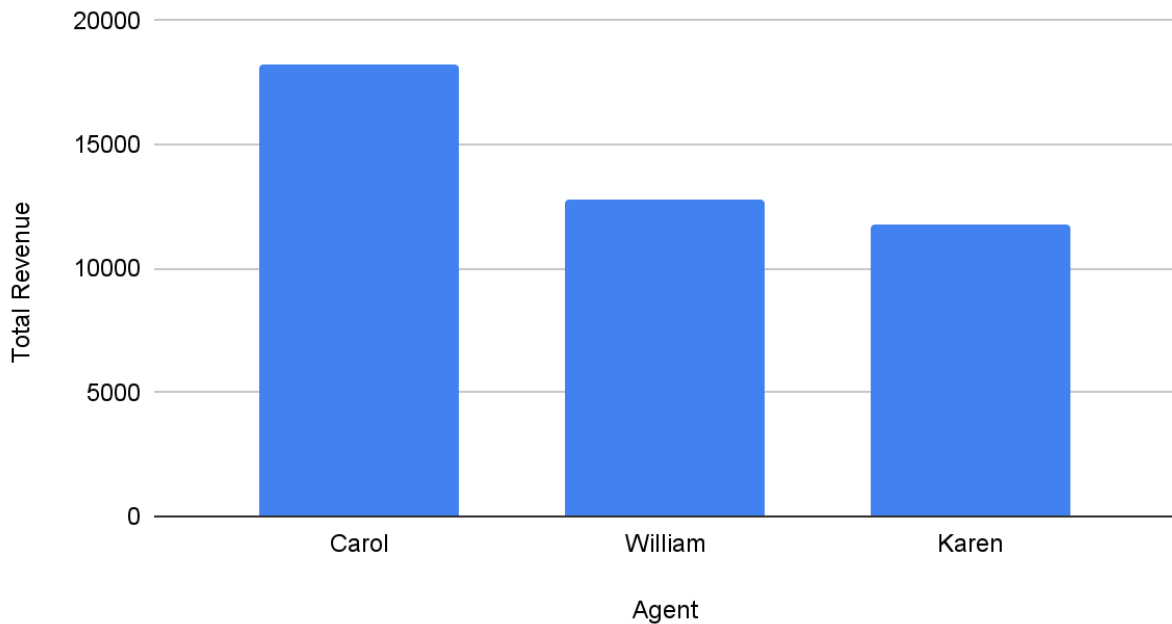


b. *Agent's performance (per-year)*
 with AgentYearRevenue as (
 select a.AgtFirstName, a.AgtLastName, extract(year from e.StartDate)
 as year, sum(e.ContractPrice) as Total_Revenue
 from Engagements e
 join Agents a on e.Agentid = a.Agentid
 group by a.AgtFirstName, a.AgtLastName, extract(year from e.StartDate)),
 Ranked as (select AgtFirstName, AgtLastName, year, Total_Revenue,
 Rank () over (partition by year order by Total_Revenue desc) as
 Agent_Performance from AgentYearRevenue)
 select AgtFirstName, AgtLastName, year, Total_Revenue, Agent_Performance
 from Ranked
 where Agent_Performance <= 3
 order by year, Agent_Performance

agtfirstname	agtlastname	year	total_revenue	agent_performance
Carol	Viescas	2017	18250	1
William	Thompson	2017	12750	2
Karen	Smith	2017	11785	3
John	Kennedy	2018	22785	1

Marianne	Wier	2018	11740	2
William	Thompson	2018	6795	3

2017 Top 3 Performing Agent

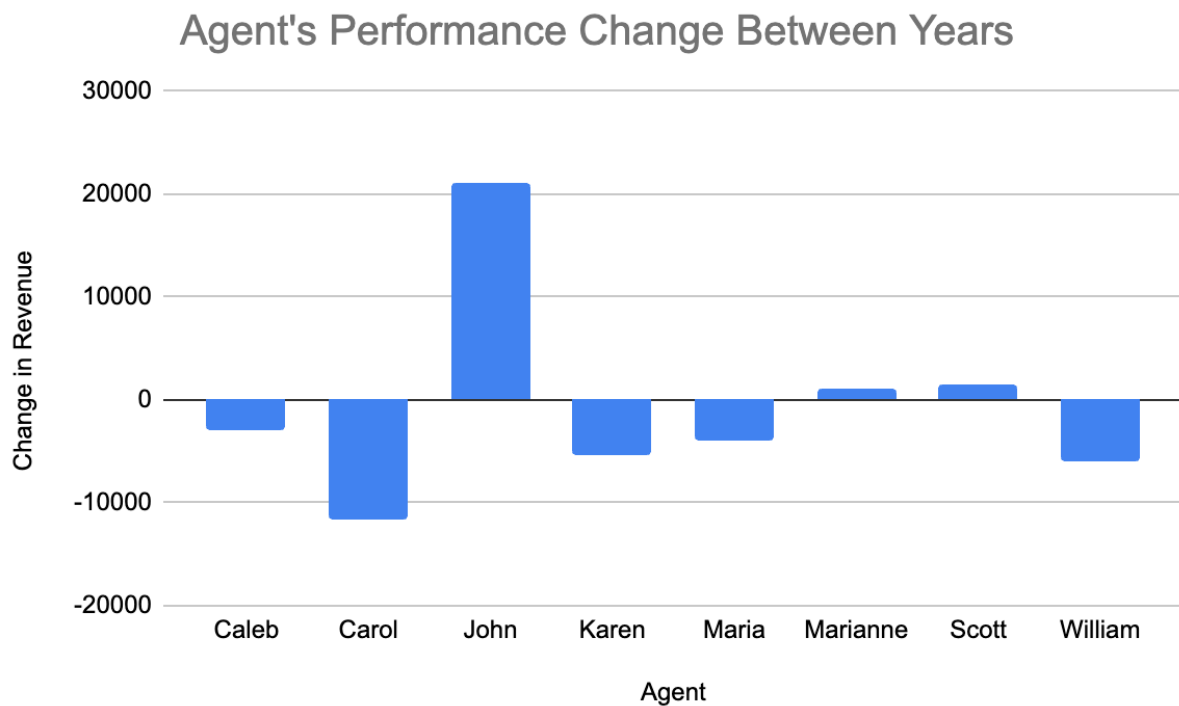


c. Calculate the change between the years

with year_sorted as (select a.AgtFirstName, a.AgtLastName, extract(year from e.StartDate) as Year, count(*) Num_Engagements, sum(ContractPrice) Total_Revenue from Engagements e
join Agents a on e.AgentID = a.AgentID
group by a.AgtFirstName, a.AgtLastName, year
order by a.agtfirstname,a.agtlastname)
Select AgtFirstName, AgtLastName, year, Num_Engagements, Total_Revenue, Total_Revenue-lag(Total_Revenue,1) over(partition by (agtfirstname, agtlastname) order by year) Change_in_Revenue from year_sorted
group by agtfirstname, agtlastname, year, Total_Revenue, num_engagements
order by agtfirstname, agtlastname

agtfirstname	agtlastname	year	num_engagements	total_revenue	change_in_revenue
Caleb	Viescas	2017	5	6760	
Caleb	Viescas	2018	3	3885	-2875
Carol	Viescas	2017	14	18250	
Carol	Viescas	2018	5	6550	-11700

John	Kennedy	2017	3	1650	
John	Kennedy	2018	9	22785	21135
Karen	Smith	2017	8	11785	
Karen	Smith	2018	8	6520	-5265
Maria	Patterson	2017	10	8435	
Maria	Patterson	2018	5	4390	-4045
Marianne	Wier	2017	9	10575	
Marianne	Wier	2018	8	11740	1165
Scott	Bishop	2017	3	2640	
Scott	Bishop	2018	3	4080	1440
William	Thompson	2017	9	12750	
William	Thompson	2018	6	6795	-5955



Discussion:

We analyzed the performance of 8 agents between September 2017 and March 2018. Carol Viescas was the most successful agent at TuneWorks: she worked on 19 bookings, which led to a total revenue of \$24800. Since the dataset included 4 months in 2017 and about 3.5 months in 2018, we decided that those time periods were comparable and made several comparisons between the two years: assessed the top performers each year and the difference in agent performance between years. Carol was the top performer in 2017. However, she had much less bookings in 2018, leading to a \$11700 drop in revenue between her bookings in 2017 and in 2018.

John Kennedy, the agent with the highest commission rate, worked on 12 bookings, which led to a total revenue of \$24435. John was the top performer in 2018. Most of the bookings he worked on were in 2018, leading to a \$21135 increase in revenue between his bookings in 2017 and in 2018.

William Thompson was one of the top three performers during both years, even though his bookings decreased in 2018.

2. Entertainers

a. *Entertainers and associated members*

```
with entertainers_c as
(select entstagename, memberid, status from entertainers e
join entertainer_members em on e.entertainerid = em.entertainerid where status=1)
select entstagename,mbrfirstname,mbrlastname from entertainers_c ec
join members m on ec.memberid = m.memberid
order by entstagename
```

entstagename	mbrfirstname	mbrlastname
Carol Peacock Trio	Sara	Sheskey
Carol Peacock Trio	Mariya	Sergienko
Caroline Coie Cuartet	Luke	Patterson
Caroline Coie Cuartet	George	Chavez
Caroline Coie Cuartet	Kendra	Bonnicksen
Coldwater Cattle Company	David	Hamilton
Coldwater Cattle Company	Suzanne	Viescas
Coldwater Cattle Company	Robert	Brown
Coldwater Cattle Company	Manuela	Seidel
Country Feeling	Joe	Rosales III
Country Feeling	Gary	Hallmark

Country Feeling	Robert	Brown
Country Feeling	Kathryn	Patterson
Jazz Persuasion	Katherine	Ehrlich
Jazz Persuasion	Angel	Kennedy
JV & the Deep Six	Jeffrey	Smith
JV & the Deep Six	David	Viescas
JV & the Deep Six	Gary	Hallmark
JV & the Deep Six	Suzanne	Viescas
JV & the Deep Six	Luke	Patterson
Modern Dance	Jeffrey	Smith
Modern Dance	Steve	Pundt
Modern Dance	Janice	Galvin
Saturday Revue	Rachel	Patterson
Saturday Revue	Susan	McLain
Saturday Revue	Caroline	Coie
Topazz	Katherine	Ehrlich

b. Groups where all members have a status of 2

```

select e.entertainerid, entstagename
from entertainer_members em
join entertainers e on em.entertainerid = e.entertainerid
group by e.entstagename, e.entertainerid
having count(*) = sum(case when status = 2 then 1 else 0 end)

```

entertainerid	entstagename
1009	Katherine Ehrlich
1012	Susan McLain
1004	Jim Glynn

1011	Julia Schnebly
------	----------------

c. *Engagements related to Groups from B*

```
select entstagename, count(engagementnumber) num_engagements from engagements e
join entertainers et on e.entertainerid = et.entertainerid
where e.entertainerid in (select e.entertainerid from entertainer_members em
join entertainers e on em.entertainerid = e.entertainerid
group by e.entstagename, e.entertainerid
having count(*) = sum(case when status = 2 then 1 else 0 end))
group by entstagename
```

entstagename	num_engagements
Jim Glynn	9
Julia Schnebly	8
Susan McLain	6

d. *Members in multiple groups*

```
select entstagename, mbrfirstname, mbrlastname from entertainers e
join entertainer_members em on e.entertainerid = em.entertainerid
join members m on em.memberid = m.memberid
where m.memberid in (select memberid from entertainer_members
group by memberid
having count(entertainerid) > 1
order by memberid)
order by m.mbrfirstname, m.mbrlastname
```

entstagename	mbrfirstname	mbrlastname
Caroline Coie Quartet	Caroline	Coie
Saturday Revue	Caroline	Coie
JV & the Deep Six	Gary	Hallmark
Country Feeling	Gary	Hallmark
Country Feeling	George	Chavez
Caroline Coie Quartet	George	Chavez
Carol Peacock Trio	Janice	Galvin
Modern Dance	Janice	Galvin

JV & the Deep Six	Jeffrey	Smith
Modern Dance	Jeffrey	Smith
Katherine Ehrlich	Katherine	Ehrlich
Topazz	Katherine	Ehrlich
Jazz Persuasion	Katherine	Ehrlich
Caroline Coie Cuartet	Kendra	Bonnicksen
Saturday Revue	Kendra	Bonnicksen
JV & the Deep Six	Luke	Patterson
Caroline Coie Cuartet	Luke	Patterson
Jazz Persuasion	Michael	Hernandez
Modern Dance	Michael	Hernandez
Topazz	Michael	Hernandez
Coldwater Cattle Company	Robert	Brown
Country Feeling	Robert	Brown
Coldwater Cattle Company	Sara	Sheskey
Carol Peacock Trio	Sara	Sheskey
Saturday Revue	Susan	McLain
Susan McLain	Susan	McLain
Coldwater Cattle Company	Suzanne	Viescas
JV & the Deep Six	Suzanne	Viescas

e. High-demand entertainers (Total number of engagements)

```

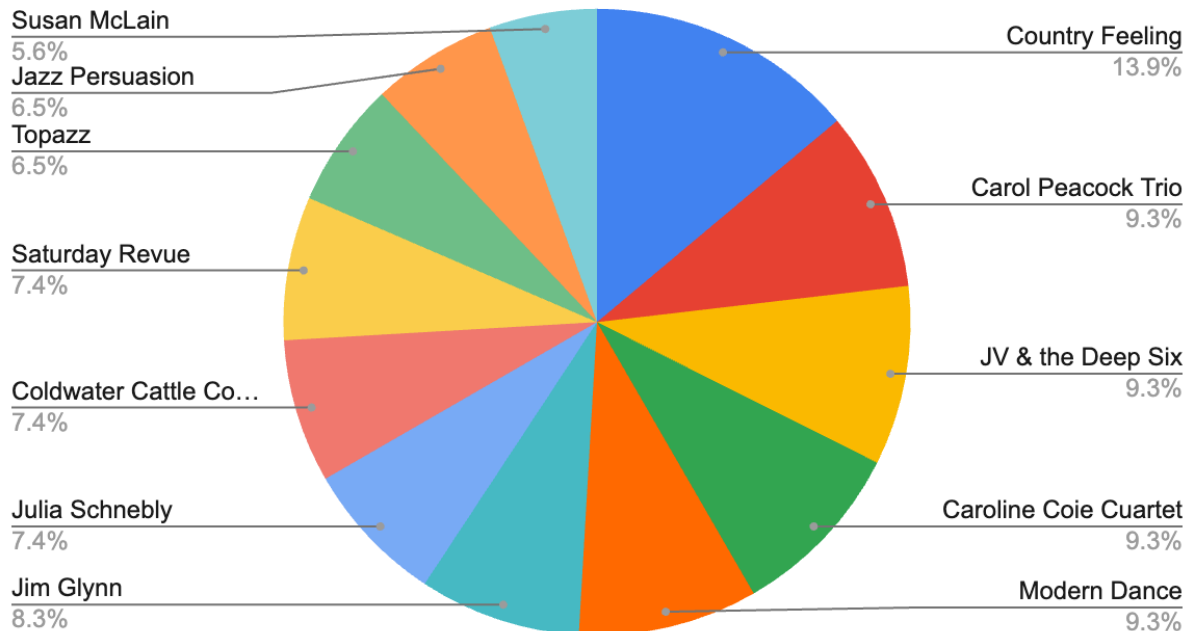
with es as (select entstageName, e.entertainerid, styleid, stylestrength from entertainers e
join entertainer_styles es on e.entertainerid = es.entertainerid),
esms as ( select entstageName, entertainerid, styleName, stylestrength from es
join musical_styles ms on es.styleid = ms.styleid)
select EntStageName, count(*) AS NumEngagements, styleName from esms
join engagements em on esms.EntertainerID = em.EntertainerID

```

where stylestrength=1
group by EntStageName, stylename
order by NumEngagements desc

entstagename	numengagements	stylename
Country Feeling	15	Country
Carol Peacock Trio	10	Show Tunes
JV & the Deep Six	10	60's Music
Caroline Coie Cuartet	10	Jazz
Modern Dance	10	Variety
Jim Glynn	9	Folk
Julia Schnebly	8	Chamber Music
Coldwater Cattle Company	8	Country Rock
Saturday Revue	8	Top 40 Hits
Topazz	7	Rhythm and Blues
Jazz Persuasion	7	Rhythm and Blues
Susan McLain	6	Folk

Number of Engagements



- f. *Who is the most popular act each year (By year)*
 with EntainerYearCounts as (select extract(year from e.StartDate) as year,
 e.Entainerid, t.EntStageName, count(*) as Num_Engagements, rank() over (partition
 by extract(year from e.StartDate) order by count(*) desc) as Top_Entainer from
 Engagements e
 join Entainers t on e.Entainerid = t.Entainerid
 group by extract(year from e.StartDate), e.EntainerID, t.EntStageName)
 select Year, EntainerID, EntStageName, Num_Engagements
 from EntainerYearCounts
 where Top_Entainer = 1
 order by Year

year	entainerid	entstagename	num_engagements
2017	1008	Country Feeling	8
2018	1008	Country Feeling	7

- g. *Groups that have no engagements*
 select t.EntainerID, t.EntStageName from Entainers as t
 left join Engagements as e on t.EntainerID = e.EntainerID
 group by t.EntainerID, t.EntStageName
 having count(e.EngagementNumber) = 0

entertainerid	entstagename
1009	Katherine Ehrlich

h. *Do higher-priced entertainers have less bookings/make less money?*

with combined_time as

(SELECT engagementnumber, entertainerid, CONCAT(startdate,' ',starttime)::timestamp as start, CONCAT(enddate,' ',stoptime)::timestamp as stop, contractprice from engagements e),pph as

(select engagementnumber, entertainerid, contractprice, (stop-start) duration, (contractprice/(SELECT EXTRACT(epoch FROM (stop-start))/3600)) as price_per_hour from combined_time)

select entstagename, count(*) as Num_Engagements, round(avg(price_per_hour),2)

average_price_per_hour, sum(contractprice) as Total_Revenue from pph

join entertainers e

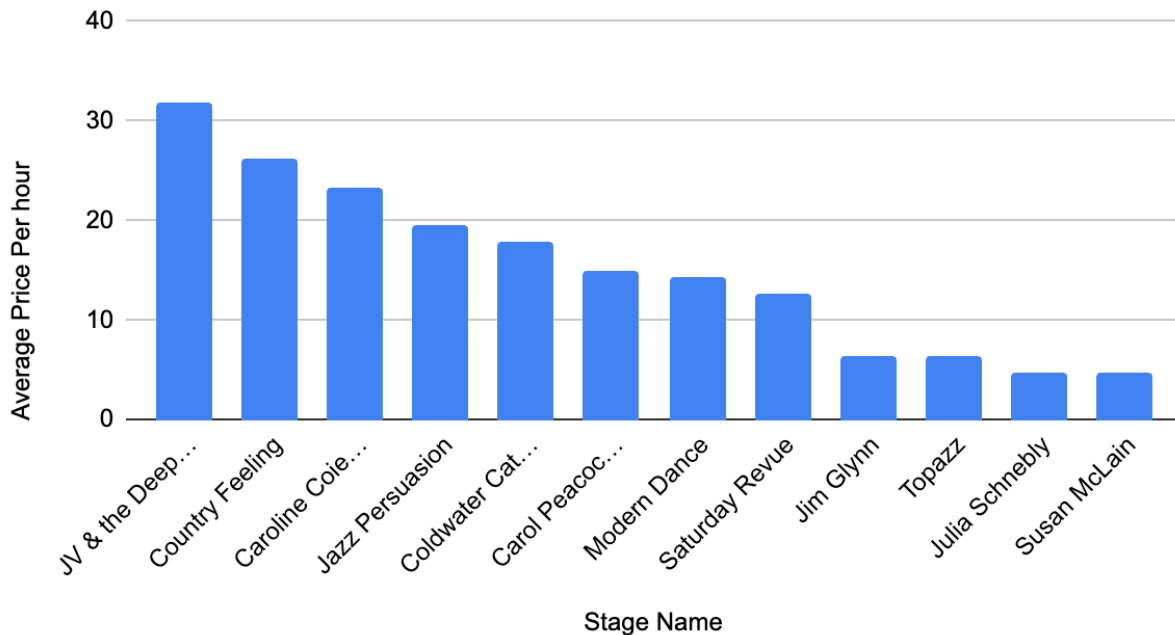
on pph.entertainerid = e.entertainerid

group by entstagename

order by average_price_per_hour desc;

entstagename	num_engagements	average_price_per_hour	total_revenue
JV & the Deep Six	10	31.81	17150
Country Feeling	15	26.15	34080
Caroline Coie Cuartet	10	23.3	14780
Jazz Persuasion	7	19.58	5480
Coldwater Cattle Company	8	17.93	14875
Carol Peacock Trio	10	14.99	10760
Modern Dance	10	14.38	14600
Saturday Revue	8	12.61	11200
Jim Glynn	9	6.39	3030
Topazz	7	6.28	6620
Julia Schnebly	8	4.68	4345
Susan McLain	6	4.66	2670

Contract Price per Hour

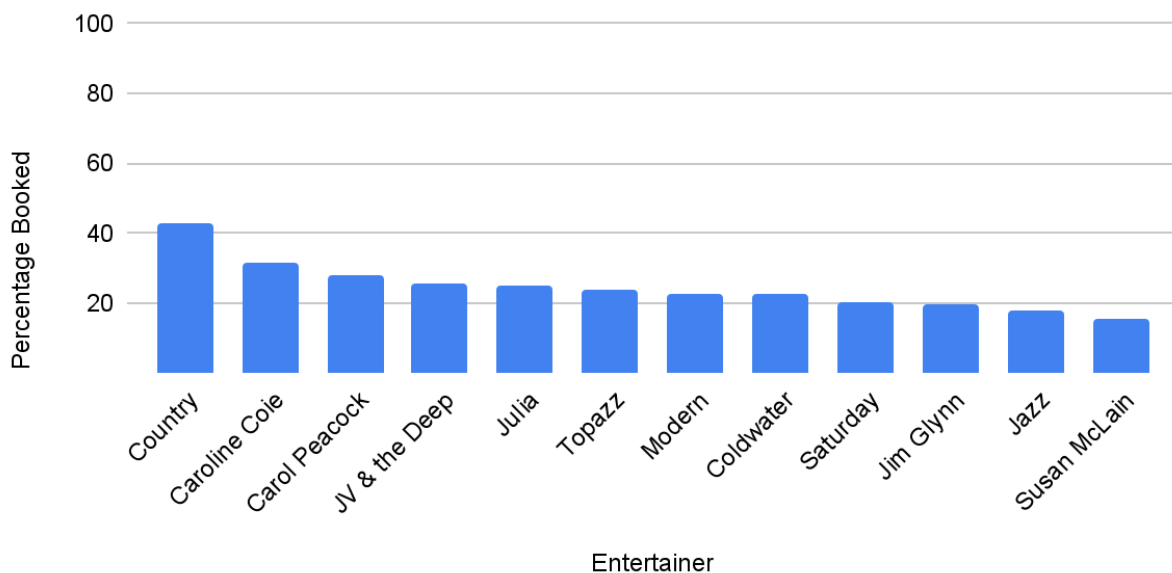


- i. *Entertainers and the percentage of days they're booked over the dataset time period*
 with combined_time as
 (select entertainerid, engagementnumber, CONCAT(startdate,' ',starttime)::timestamp as start, CONCAT(enddate,' ',stoptime)::timestamp as stop from engagements e),
 combined_time_es as
 (select entertainerid, engagementnumber, (stop-start) days_booked from combined_time),
 total as (select entstagename, (extract(days from sum(days_booked))+(extract(hours from sum(days_booked))/24)) total_days_booked from combined_time_es
 join entertainers et on combined_time_es.entertainerid = et.entertainerid
 group by entstagename)
 select entstagename, round((((total_days_booked*100)/(select extract(days from (max(stop)-min(start))) from combined_time)),2) percentage_booked from total
 order by percentage_booked desc

entstagename	percentage_booked
Country Feeling	43.06
Caroline Coie Cuartet	31.55
Carol Peacock Trio	27.84
JV & the Deep Six	25.72

Julia Schnebly	24.78
Topazz	24.07
Modern Dance	22.83
Coldwater Cattle Company	22.57
Saturday Revue	20.44
Jim Glynn	19.49
Jazz Persuasion	18.16
Susan McLain	15.69

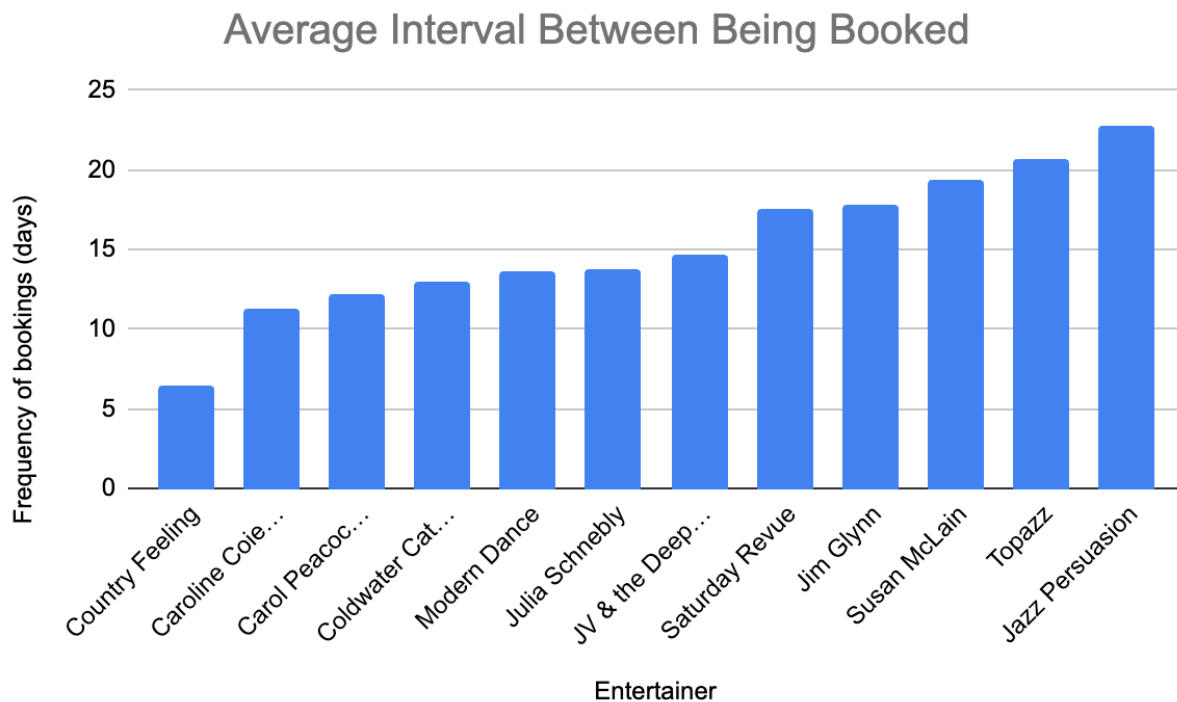
Entertainers and the percentage of days they're booked over the dataset time period



j. *Avg interval between being booked*
with combined_time as
(SELECT entertainerid, engagementnumber, CONCAT(startdate,' ',starttime)::timestamp
as start, CONCAT(enddate,' ',stoptime)::timestamp as stop from engagements e),
intervals as
(select entertainerid, start-lag(stop,1) over(partition by entertainerid order by start)
intervals from combined_time)
select entstagename, round((extract(day from avg(intervals)))+(extract(hour from
avg(intervals))/24),2) frequency_of_bookings from intervals

join entertainers et on intervals.entertainerid = et.entertainerid
group by entstagename
order by frequency_of_bookings asc

entstagename	Frequency_of_bookings (days)
Country Feeling	6.46
Caroline Coie Cuartet	11.25
Carol Peacock Trio	12.17
Coldwater Cattle Company	13.04
Modern Dance	13.67
Julia Schnebly	13.79
JV & the Deep Six	14.71
Saturday Revue	17.5
Jim Glynn	17.83
Susan McLain	19.42
Topazz	20.63
Jazz Persuasion	22.75



Discussion:

Our initial analysis was to determine which members are part of each of the thirteen entertainers under TuneWorks. The entertainers are a mix of groups and solo acts. We made the assumption that 2 meant members who were inactive; there were several solo acts who were potentially inactive (Katherine Ehrlich, Susan McLain, Jim Glynn, Julia Schnebly). Katherine Ehrlich was not able to get bookings as a solo act, although she has been part of successful duos. The other three were booked a total of 23 times before they potentially became inactive. Additionally, we interestingly discovered that many members are in multiple groups.

Country Feeling was the most booked act (15 bookings) and most frequently booked act, and Susan McLain was the least booked act (6 bookings). Further investigation showed that Country Feeling was the most booked act in 2017 and 2018, and their bookings were most likely for longer periods as they were booked for almost half of the data set time period. Jazz Persuasion was the least frequently booked act.

Four different acts had 10 bookings, and they specialized in four different styles that were not Country. Thus, there was not a prominent style associated with these five popular acts.

Interestingly, two agents and two members have the last name Viescas. As Viescas does not seem as common of a last name, it could be a full family employed by TuneWorks.

3. Customers

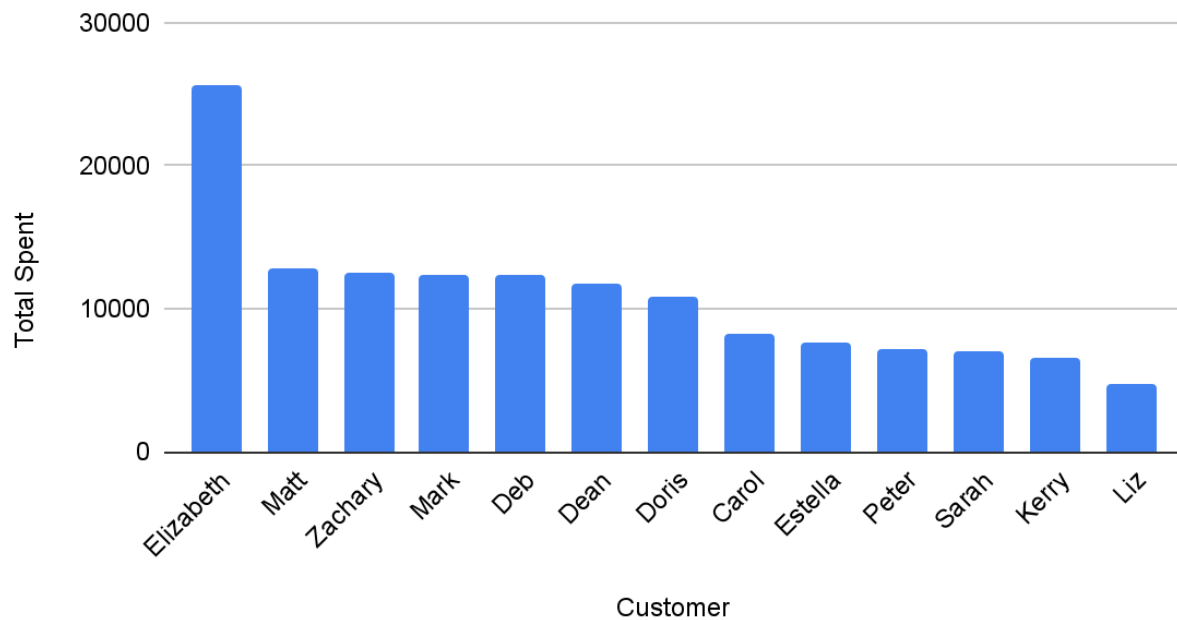
a. Customers that spend the most

```
select c.custfirstname, c.custlastname, sum(e.contractprice) as total_spent, rank()
over(order by sum(e.contractprice) desc) top_spenders from engagements e
join customers c on e.customerid = c.customerid
```

group by c.customerid, c.custfirstname, c.custlastname
order by total_spent desc

custfirstname	custlastname	total_spent	top_spenders
Elizabeth	Hallmark	25585	1
Matt	Berg	12850	2
Zachary	Ehrlich	12455	3
Mark	Rosales	12420	4
Deb	Waldal	12320	5
Dean	McCrae	11800	6
Doris	Hartwig	10795	7
Carol	Viescas	8255	8
Estella	Pundt	7560	9
Peter	Brehm	7250	10
Sarah	Thompson	7090	11
Kerry	Patterson	6525	12
Liz	Keyser	4685	13

Customers That Spend The Most



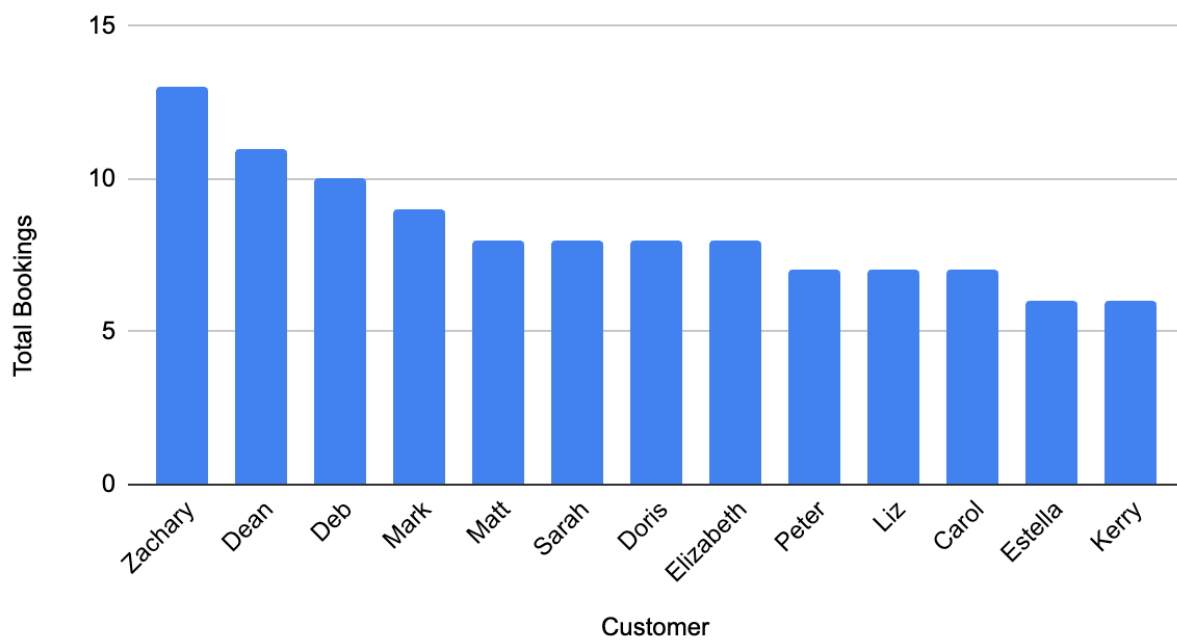
b. *Customers makes the most number of bookings*

```
select c.custfirstname, c.custlastname, count(*) as total_bookings, rank() over(order by
count(*) desc) Highest_bookings from engagements e
join customers c on e.customerid = c.customerid
group by c.customerid, c.custfirstname, c.custlastname
order by Highest_bookings
```

custfirstname	custlastname	total_bookings	highest_bookings
Zachary	Ehrlich	13	1
Dean	McCrae	11	2
Deb	Waldal	10	3
Mark	Rosales	9	4
Matt	Berg	8	5

Sarah	Thompson	8	5
Doris	Hartwig	8	5
Elizabeth	Hallmark	8	5
Peter	Brehm	7	9
Liz	Keyser	7	9
Carol	Viescas	7	9
Estella	Pundt	6	12
Kerry	Patterson	6	12

Customers Makes The Most Number of Bookings



c. *Customers who make frequent bookings*

```

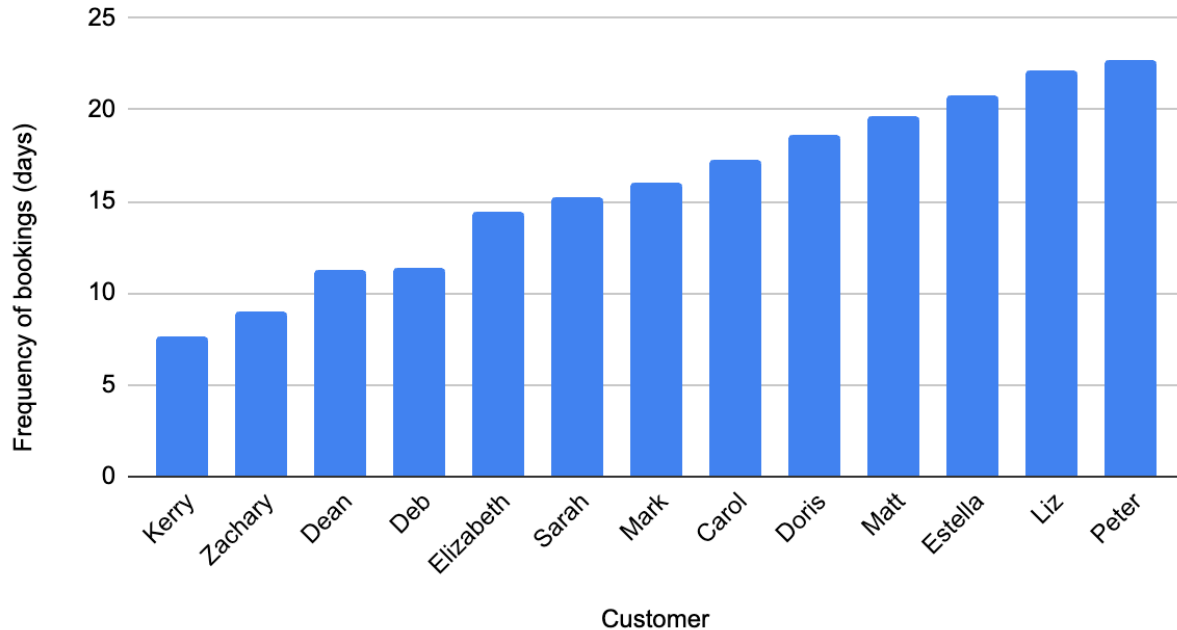
with combined_time as (select customerid, engagementnumber, concat(startdate,'
',starttime)::timestamp as start, concat(enddate,' ',stoptime)::timestamp as stop from
engagements e),
intervals as (select customerid, start-lag(stop,1) over(partition by customerid order by
start) intervals from combined_time)
select c.custfirstname, c.custlastname, round((extract(day from
avg(intervals)))+(extract(hour from avg(intervals))/24),2) frequency_of_bookings from
intervals
join customers c
on intervals.customerid = c.customerid
group by c.custfirstname, c.custlastname
order by frequency_of_bookings asc

```

custfirstname	custlastname	Frequency_of_bookings (days)
Kerry	Patterson	7.58
Zachary	Ehrlich	9
Dean	McCrae	11.25
Deb	Waldal	11.38
Elizabeth	Hallmark	14.46
Sarah	Thompson	15.25
Mark	Rosales	16.04
Carol	Viescas	17.33
Doris	Hartwig	18.67
Matt	Berg	19.67
Estella	Pundt	20.83
Liz	Keyser	22.13

Peter	Brehm	22.67
-------	-------	-------

Customers Who Make The Most Frequent Bookings (Days)



d. *Entertainers and customers with the same address*

```
select entstagename, entstreetaddress, entcity, entzipcode, custfirstname, custlastname,
       custstreetaddress, custcity, custzipcode from entertainers e
```

```
join customers c on e.entstreetaddress = c.custstreetaddress and e.entcity = c.custcity
```

entstagename	entstreetaddress	entcity	entzipcode	custfirstname	custlastname	custstreetaddress	custcity	custzipcode
Coldwater Cattle Company	4726 - 11th Ave. N.E.	Seattle	98105	Doris	Hartwig	4726 - 11th Ave. N.E.	Seattle	98105
Carol Peacock Trio	4110 Old Redmond Rd.	Redmond	98052	Dean	McCrae	4110 Old Redmond Rd.	Redmond	98052
Jim Glynn	13920 S.E. 40th Street	Belle vue	98009	Liz	Keyser	13920 S.E. 40th Street	Belle vue	98006

Discussion:

Elizabeth Hallmark was the highest-spending customer: the total amount spent was \$25,585. This indicates strong purchasing power and high engagement. Five of the top spenders after Elizabeth spent around \$12000, which was half of what Elizabeth spent. Therefore, establishing a high-spender VIP tier could be viable. Zachary Ehrlich made the most bookings, but Kerry Patterson made the most frequent bookings.

Additionally, it was noted that certain entertainers shared the same address as customers. There are several potential explanations: the address may be an apartment complex. Another explanation is that the customer has previous connections to the entertainer.

4. Musical Styles

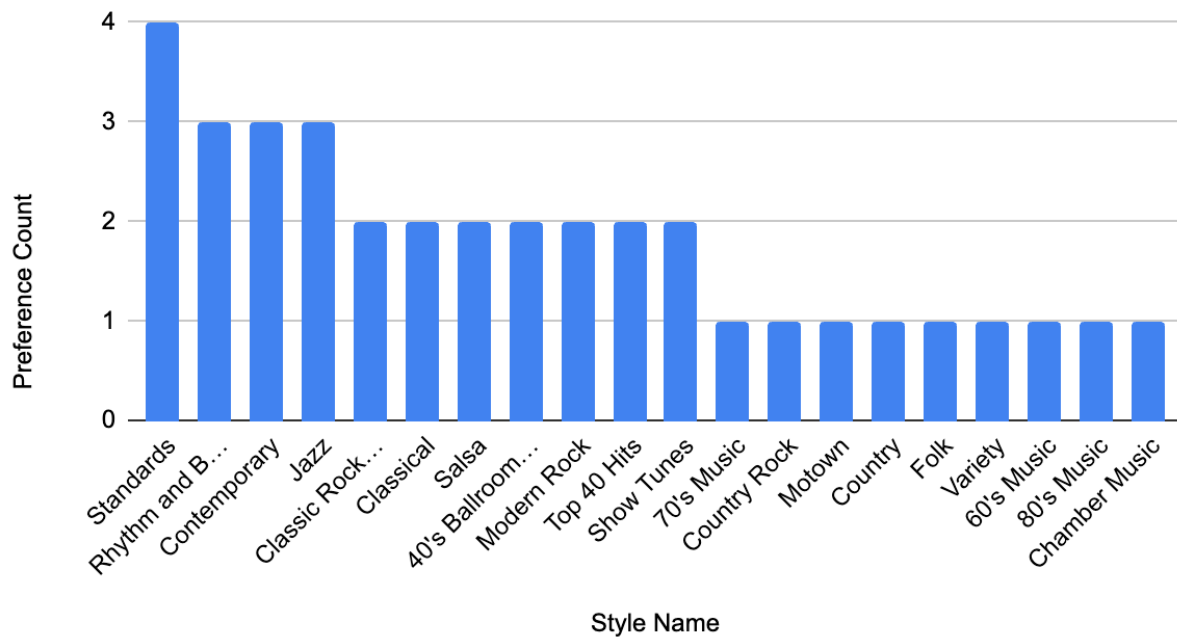
a. *Most popular musical styles with customers*

```
select ms.stylename, count(*) as preference_count from musical_preferences mp
join musical_styles ms on mp.styleid = ms.styleid
group by ms.stylename
order by preference_count desc
```

stylename	preference_count
Standards	4
Rhythm and Blues	3
Contemporary	3
Jazz	3
Classic Rock & Roll	2
Classical	2
Salsa	2
40's Ballroom Music	2
Modern Rock	2
Top 40 Hits	2
Show Tunes	2
70's Music	1
Country Rock	1
Motown	1
Country	1

Folk	1
Variety	1
60's Music	1
80's Music	1
Chamber Music	1

Most Popular Musical Styles with Customers



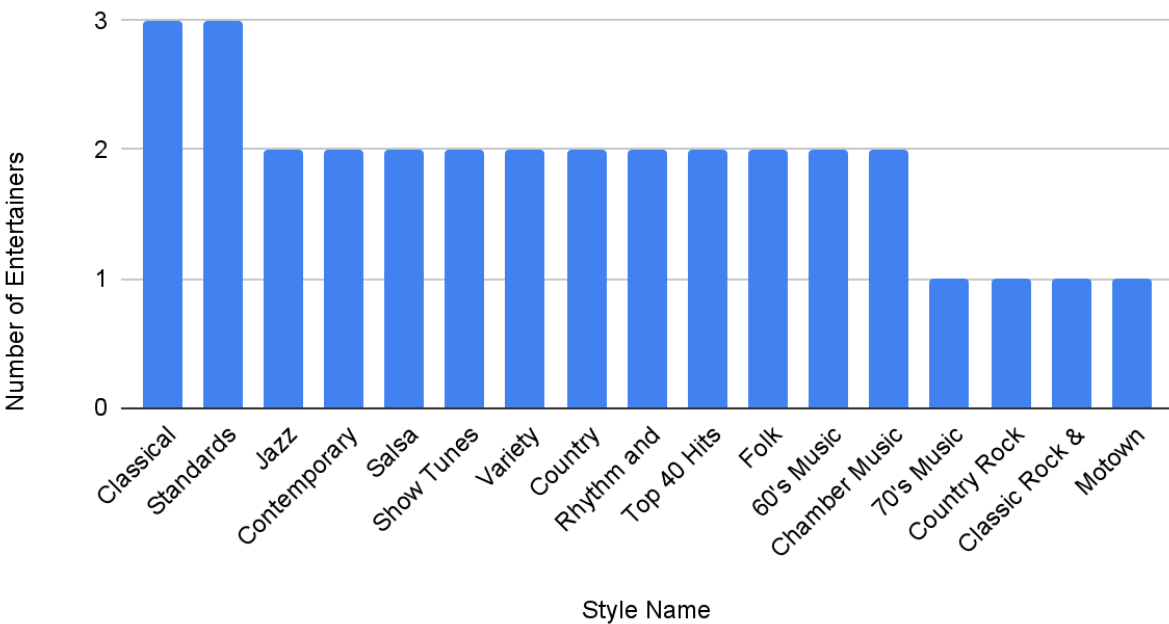
b. Which musical styles are most covered by the current entertainers?

```
select ms.stylename, count(*) as num_entertainers from entertainer_styles es
join musical_styles ms on es.styleid = ms.styleid
group by ms.stylename
order by num_entertainers desc
```

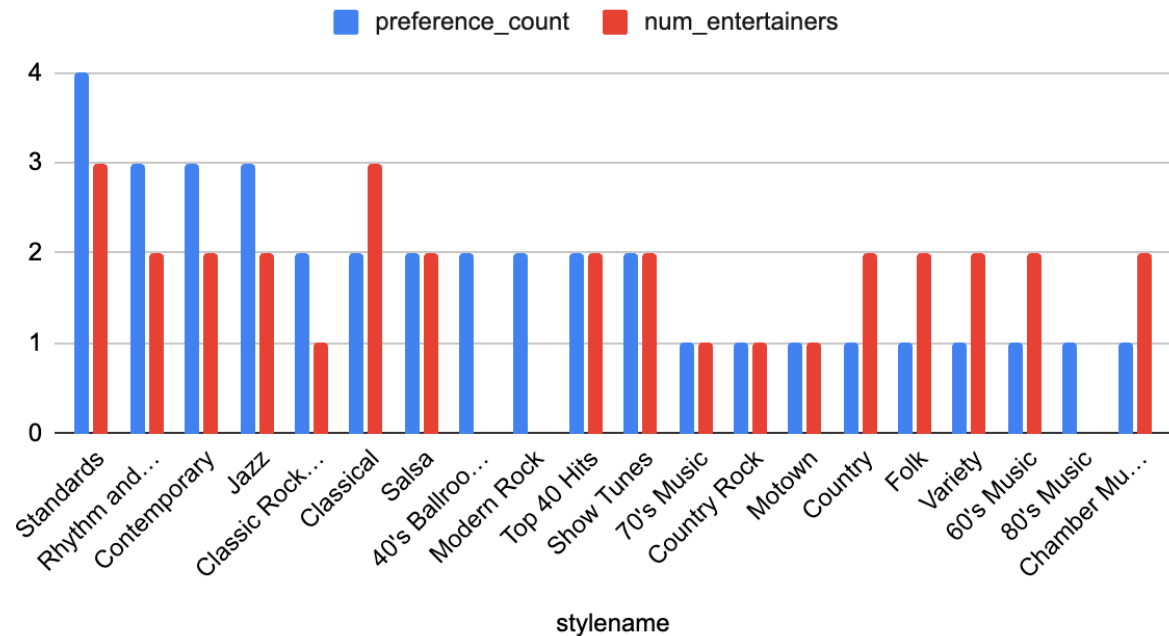
stylename	num_entertainers
Classical	3
Standards	3
Jazz	2

Contemporary	2
Salsa	2
Show Tunes	2
Variety	2
Country	2
Rhythm and Blues	2
Top 40 Hits	2
Folk	2
60's Music	2
Chamber Music	2
70's Music	1
Country Rock	1
Classic Rock & Roll	1
Motown	1

Musical Styles Covered by the Current Entertainers



Music Style Preference Count v.s. Number of Entertainers



c. All musical styles associated with each entertainer

with entertainer as (select entstageName, styleid, stylestrength from entertainers e
join entertainer_styles es on e.entertainerid = es.entertainerid)

```

select entstagename, ms.stylename, stylestrength from entertainer e2
join musical_styles ms on e2.styleid = ms.styleid
group by ms.stylename, entstagename, stylestrength
order by entstagename, stylestrength

```

entstagename	stylename	stylestrength
Carol Peacock Trio	Show Tunes	1
Carol Peacock Trio	Contemporary	2
Carol Peacock Trio	Standards	3
Caroline Coie Cuartet	Jazz	1
Caroline Coie Cuartet	Contemporary	2
Coldwater Cattle Company	Country Rock	1
Coldwater Cattle Company	Country	2
Country Feeling	Country	1
Country Feeling	60's Music	2
Jazz Persuasion	Rhythm and Blues	1
Jazz Persuasion	Salsa	2
Jazz Persuasion	Jazz	3
Jim Glynn	Folk	1
Julia Schnebly	Chamber Music	1
Julia Schnebly	Classical	2
Julia Schnebly	Show Tunes	3
JV & the Deep Six	60's Music	1
JV & the Deep Six	Classic Rock & Roll	2
Katherine Ehrlich	Chamber Music	1
Katherine Ehrlich	Classical	2
Katherine Ehrlich	Standards	3

Modern Dance	Variety	1
Modern Dance	Salsa	2
Modern Dance	Top 40 Hits	3
Saturday Revue	Top 40 Hits	1
Saturday Revue	70's Music	2
Saturday Revue	Standards	3
Susan McLain	Folk	1
Susan McLain	Classical	2
Topazz	Rhythm and Blues	1
Topazz	Motown	2
Topazz	Variety	3

Discussion:

The most preferred musical style among customers is Standards, with a total of 4 customer selections. This indicates that traditional “Standards” music has strong appeal across a meaningful portion of the customer base. Rhythm and Blues, Jazz, and Contemporary were also popular.

The most represented styles are Classical and Standards, each with 3 entertainers. However, Standard is the weakest style for those 3 entertainers. Several styles (Jazz, Contemporary, Salsa, Show Tunes, Variety, Country, Rhythm and Blues, Top 40 Hits, Folk, 60’s Music, and Chamber Music) are moderately represented, each with 2 entertainers. A smaller group of styles, such as 70’s Music, Country Rock, Classic Rock & Roll, and Motown, have 1 entertainer each.

Thus, the overlap is poor between the preferred musical styles and the main musical style strengths of the entertainers under TuneWorks. While Topazz and Jazz Persuasion specializes in Rhythm and Blues, only Caroline Coie Quartet specializes in Jazz and no entertainer specializes in Contemporary. The top group, Country Feeling, specializes in Country, which only one customer prefers.

5. Engagements

a. Average length of engagements

with combined_time as

(selet engagementnumber, concat(startdate,' ',starttime)::timestamp as start,

concat(enddate,' ',stoptime)::timestamp as stop from engagements e)

select avg(stop-start) average_engagement_length from combined_time

average_engagement_length

5 days 06:22:46.666667

b. *Number of engagements per month (based it on startdate)*

with months as (

select case

when monthnumber=1 then 'January'

when monthnumber=2 then 'February'

when monthnumber=3 then 'March'

when monthnumber=4 then 'April'

when monthnumber=5 then 'May'

when monthnumber=6 then 'June'

when monthnumber=7 then 'July'

when monthnumber=8 then 'August'

when monthnumber=9 then 'September'

when monthnumber=10 then 'October'

when monthnumber=11 then 'November'

when monthnumber=12 then 'December'

end as month, engagementnumber from engagements e

join ztblmonths z on extract(month from e.startdate)=z.monthnumber and extract(year from e.startdate)=z.yearnumber)

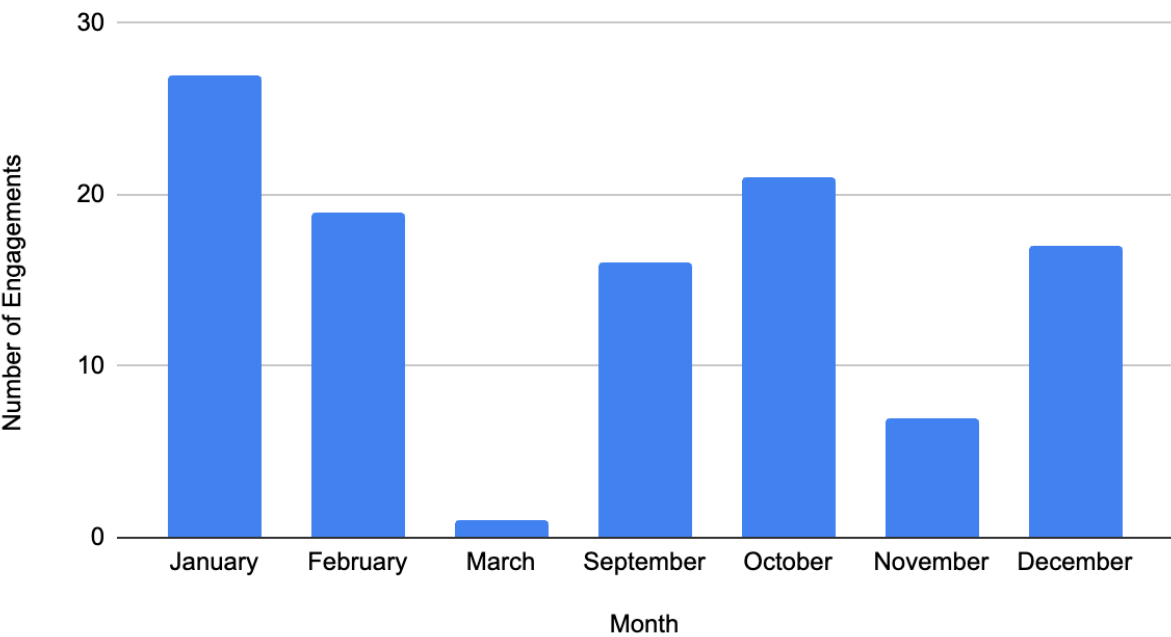
select month, count(engagementnumber) num_of_engagements from months

group by month

order by num_of_engagements desc

month	num_of_engagements
January	27
October	21
February	19
December	17
September	16
November	7
March	1

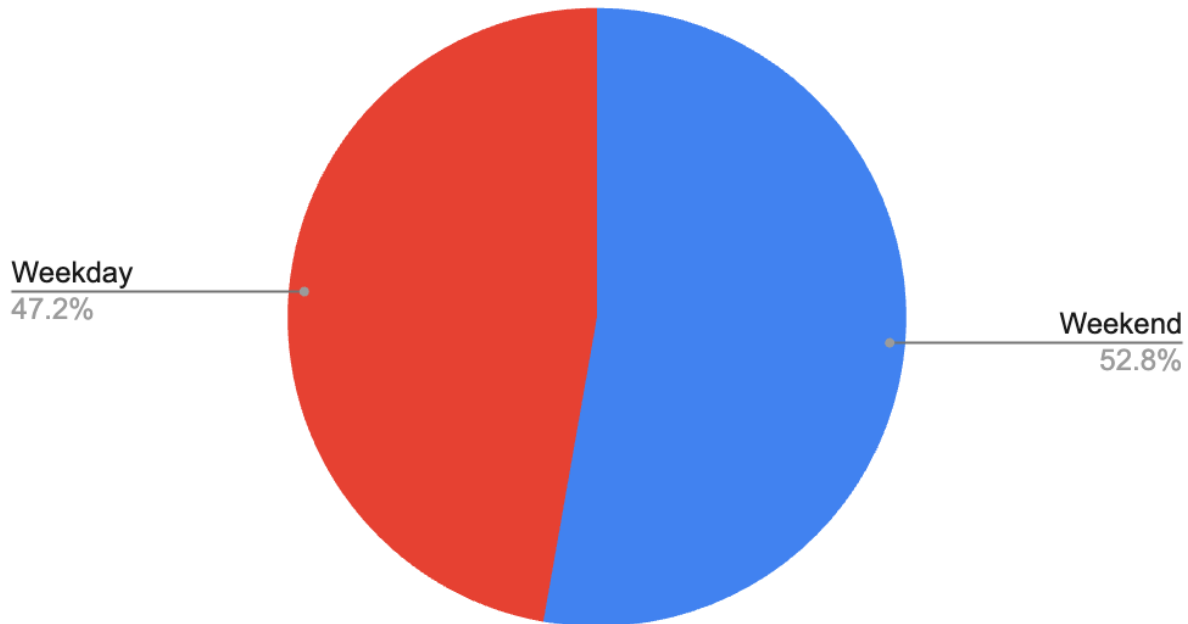
Number of Engagements Per Month



c. *When do the engagements start? (Weekdays/weekends)*
with counts as (select count(case when (e.startdate = z.weekstart) or (e.startdate = z.weekend) then 1 end) as num_weekend_engagements, count(*) as num_of_engagements from engagements e left join ztblweeks z on (e.startdate = z.weekstart) or (e.startdate = z.weekend))
select
num_weekend_engagements,round((num_weekend_engagements*100/num_of_engagements)) percentage_of_weekend_engagements,
(num_of_engagements-num_weekend_engagements) num_weekday_engagements,
round(((num_of_engagements-num_weekend_engagements)*100/num_of_engagements)) percentage_of_weekday_engagements, num_of_engagements from counts

num_weekend_engagements	percentage_of_weekend_engagements	num_weekday_engagements	percentage_of_weekday_engagements	num_of_engagements
57	52	51	47	108

Engagement Start Date (Weekdays/weekends)



Discussion:

The engagements on average seem to be about 5 days. During this seven month time span, the most popular month for bookings was January, as there were 27 bookings. October and February seemed to be also popular: there were 21 bookings in October and 19 bookings in February. It is difficult to establish a trend, besides people potentially celebrating the new year.

While more bookings start on the weekend, it is only 5% more frequent than bookings that start on the weekday.

6. Joining

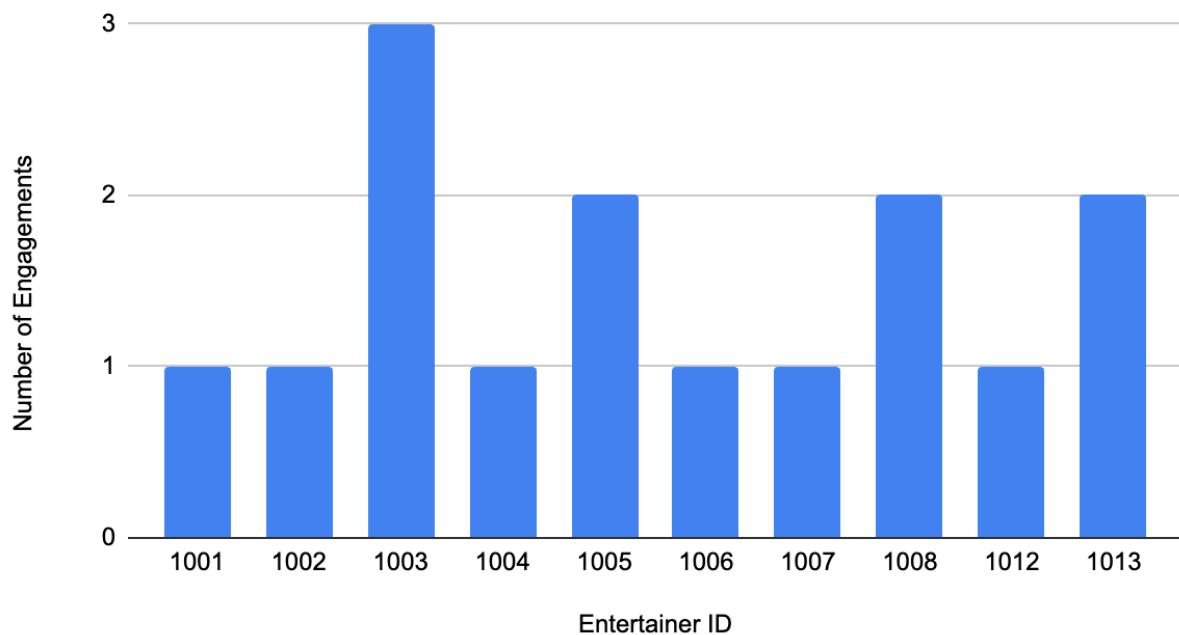
a. Which agents and entertainers work the most together

```
with pairing as (select agentid, entertainerid from engagements order by agentid),
ranking as (
select agentid, entertainerid, count(entertainerid) num_of_engagements,
rank()over(partition by agentid order by count(entertainerid) desc) top_partner from
pairing
group by agentid, entertainerid
order by agentid, num_of_engagements desc)
select a.agtfirstname, a.agtlastname, e.entstagename, num_of_engagements,top_partner
from ranking r
join agents a
on r.agentid = a.agentid
join entertainers e
on r.entertainerid = e.entertainerid
where top_partner=1
order by agtfirstname,agtlastname, num_of_engagements desc
```

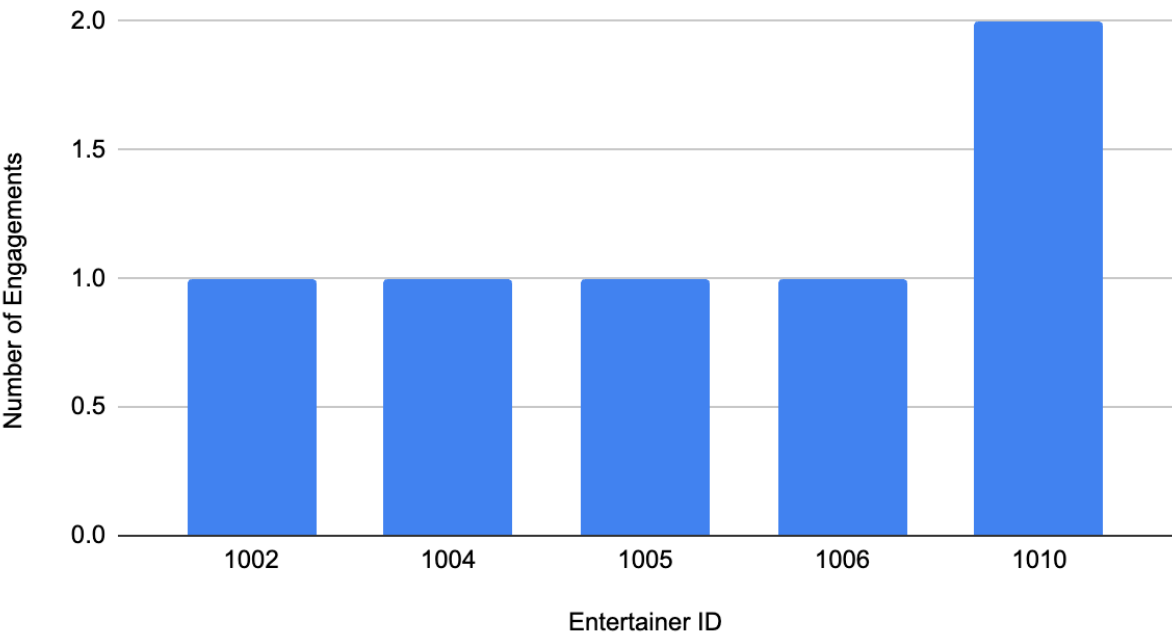
agtfirstname	agtlastname	entstagename	num_of_engagements	top_partner
--------------	-------------	--------------	--------------------	-------------

Caleb	Viescas	Country Feeling	3	1
Carol	Viescas	Country Feeling	3	1
Carol	Viescas	Modern Dance	3	1
John	Kennedy	Country Feeling	3	1
Karen	Smith	Julia Schnebly	3	1
Karen	Smith	Country Feeling	3	1
Maria	Patterson	Carol Peacock Trio	4	1
Marianne	Wier	Carol Peacock Trio	3	1
Marianne	Wier	Caroline Coie Cuartet	3	1
Scott	Bishop	Saturday Revue	2	1
William	Thompson	JV & the Deep Six	3	1

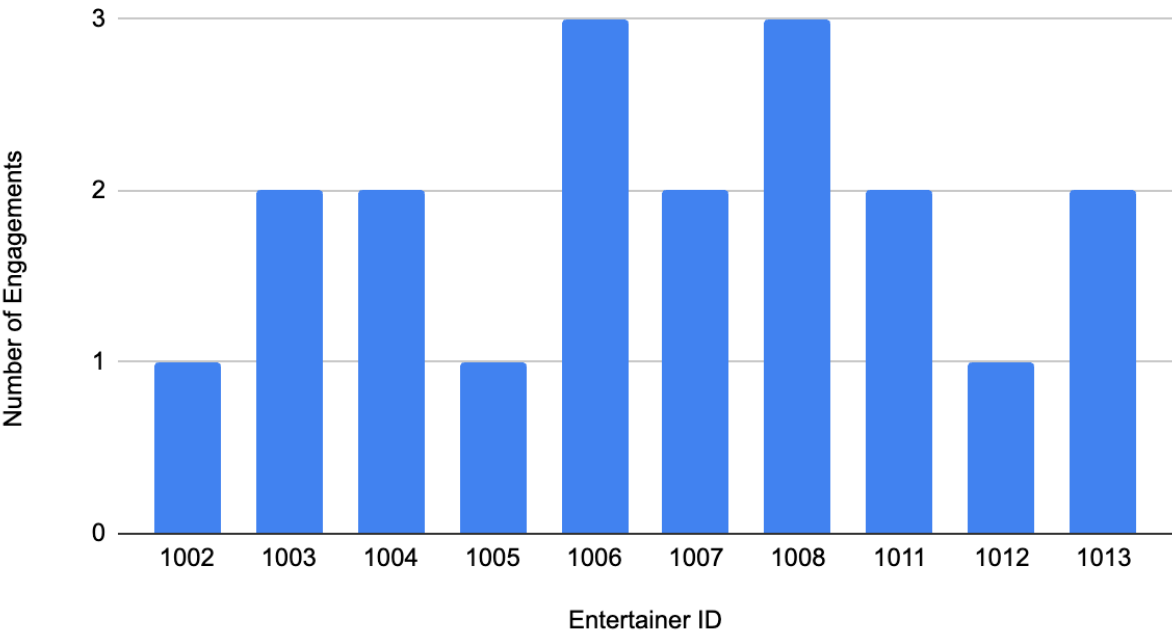
Agent 1's Engagements with Different Entertainers



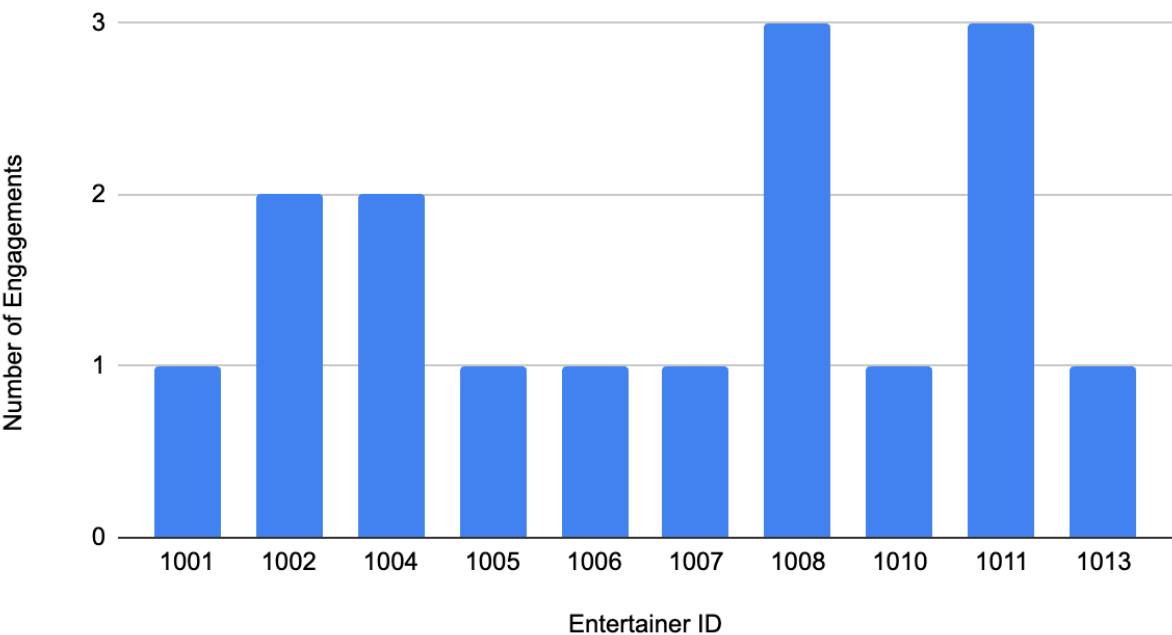
Agent 2's Engagements with Different Entertainers



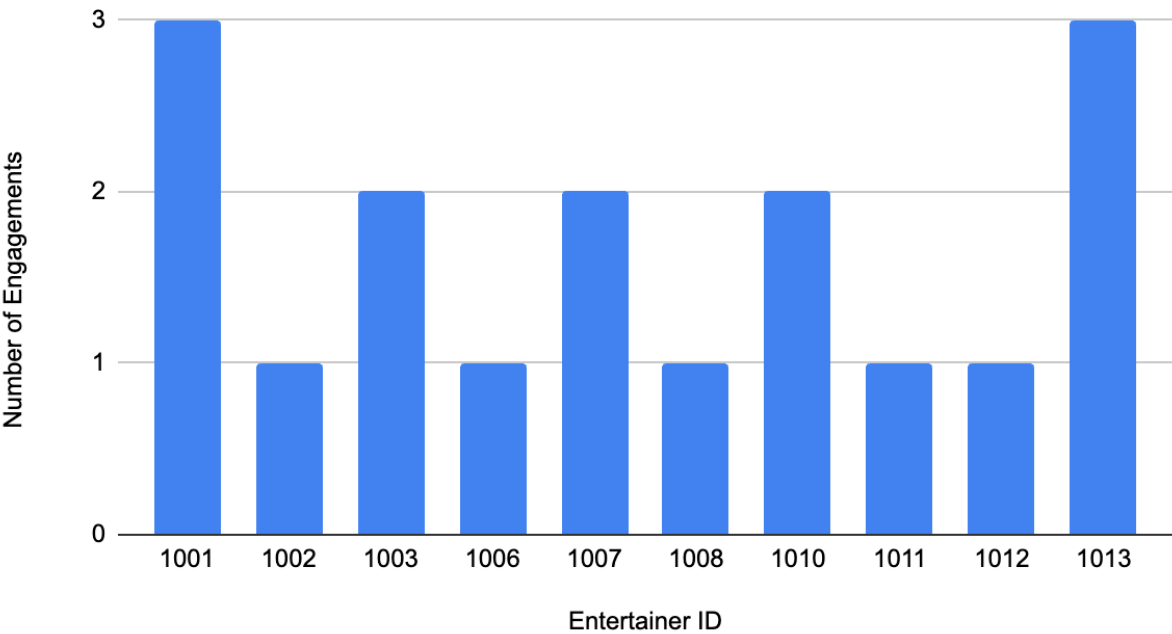
Agent 3's Engagements with Different Entertainers



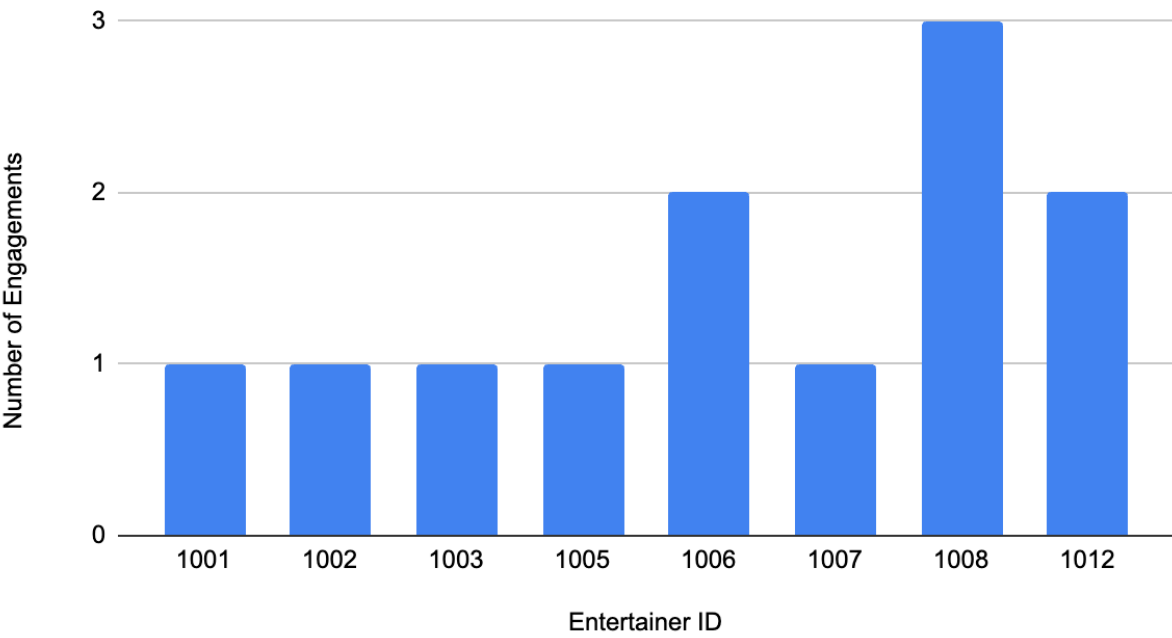
Agent 4's Engagements with Different Entertainers



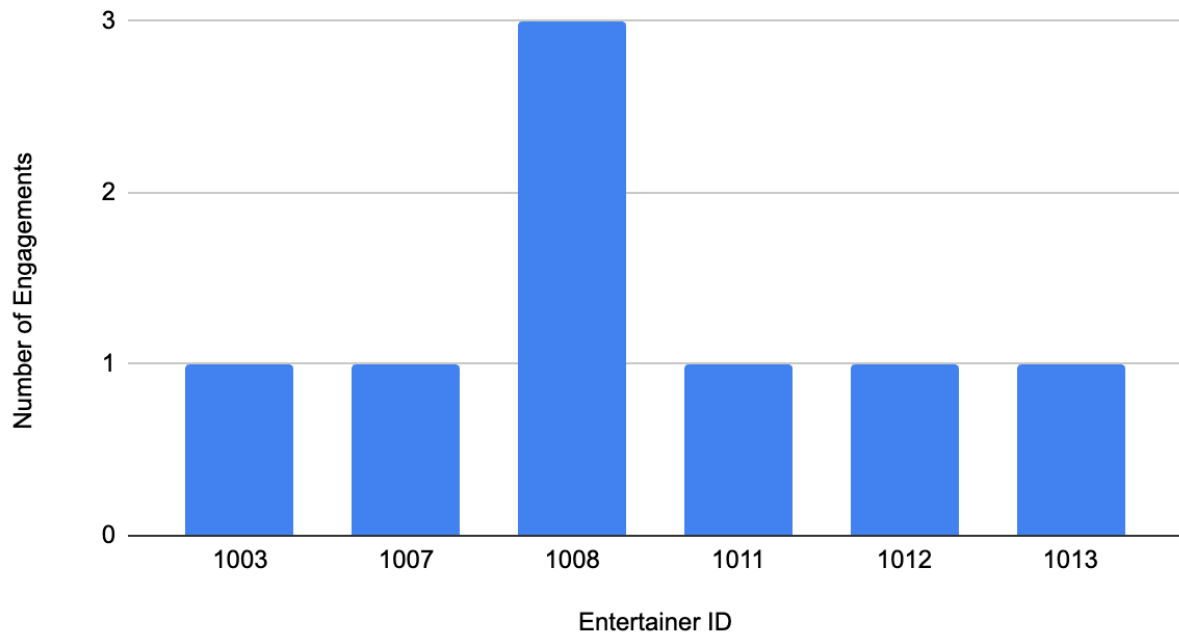
Agent 5's Engagements with Different Entertainers



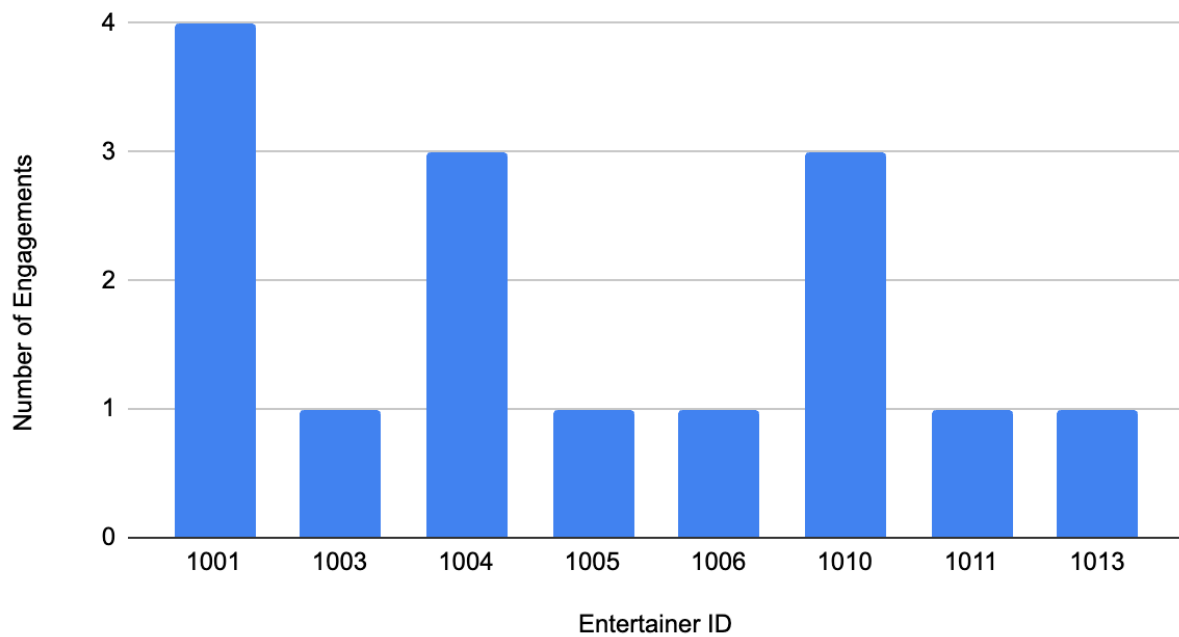
Agent 6's Engagements with Different Entertainers



Agent 7's Engagements with Different Entertainers



Agent 8's Engagements with Different Entertainers



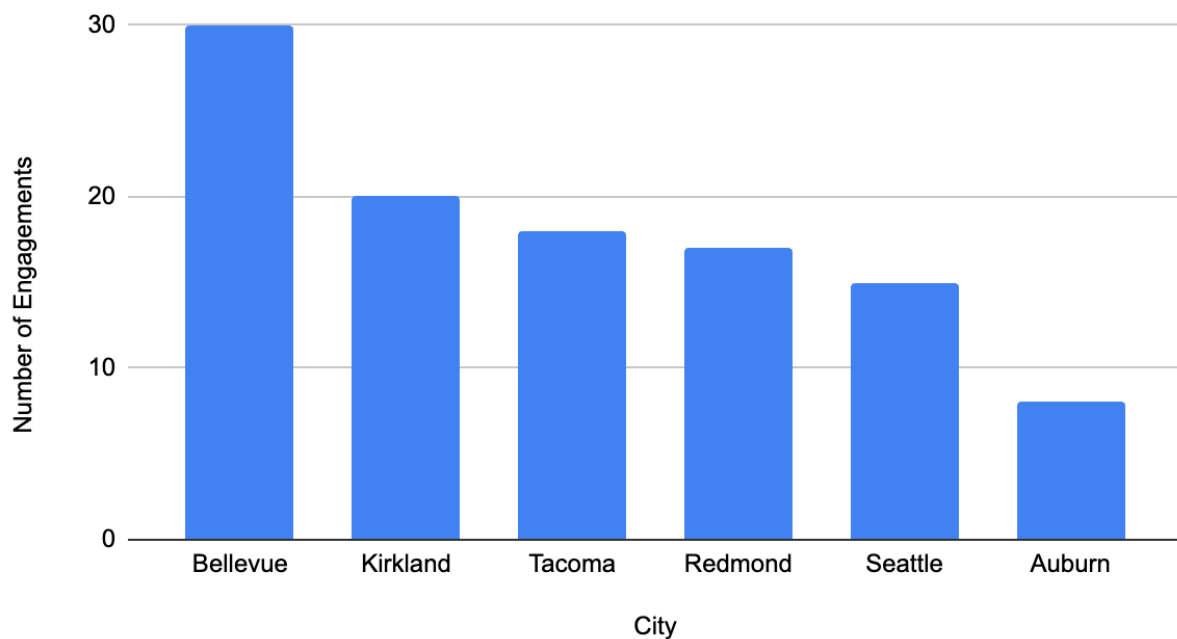
b. *Active cities for customers/engagements*

```
select custcity, count(c.custcity) num_of_engagements from engagements e
join customers c on e.customerid = c.customerid
```

group by custcity
order by num_of_engagements desc

custcity	num_of_engagements
Bellevue	30
Kirkland	20
Tacoma	18
Redmond	17
Seattle	15
Auburn	8

Active Cities for Customers/Engagements



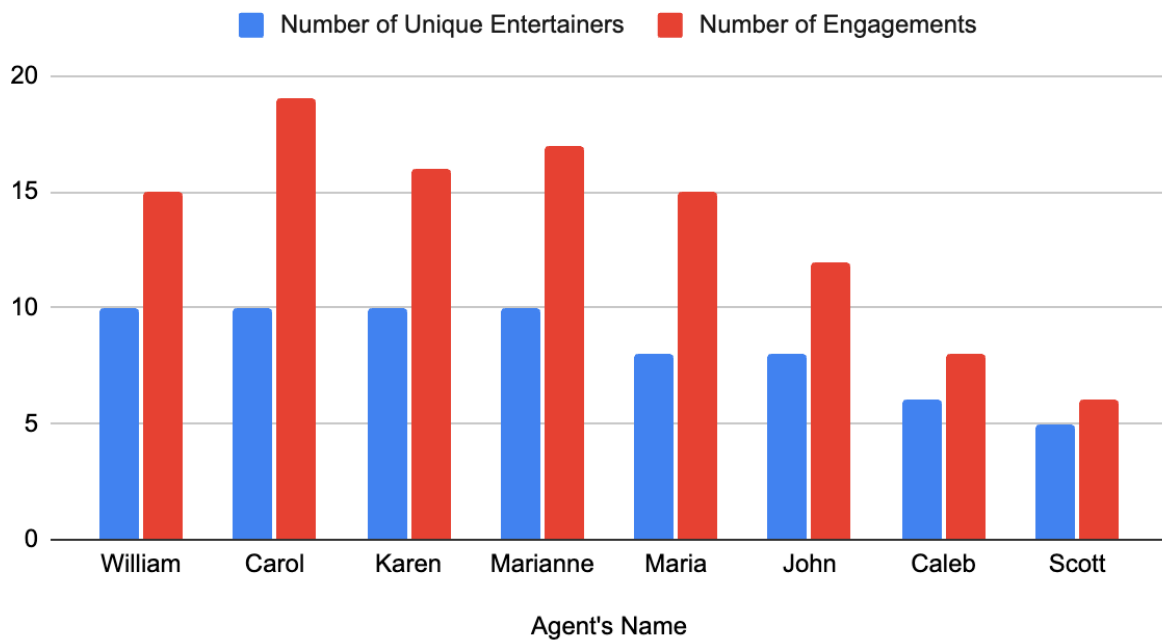
c. *Are agents diversified in entertainers or relying on a few?*

```
select e.Agentid, a.AgtFirstName, a.AgtLastName, count(distinct e.Entertainerid) as
Num_Unique_Entertainers, count(*) as Num_Engagements from Engagements e
join Agents a on e.Agentid = a.Agentid
group by e.Agentid, a.AgtFirstName, a.AgtLastName
order by NumDistinctEntertainers desc
```

agentid	agtfirstname	agtlastname	num_unique_entertainers	num_engagements
---------	--------------	-------------	-------------------------	-----------------

1	William	Thompson	10	15
3	Carol	Viescas	10	19
4	Karen	Smith	10	16
5	Marianne	Wier	10	17
8	Maria	Patterson	8	15
6	John	Kennedy	8	12
7	Caleb	Viescas	6	8
2	Scott	Bishop	5	6

Agents / Entertainers Collaboration Diversity



Discussion:

Country Feeling is the client that four different agents have worked most often with. One of those four agents, Karen Smith, has also worked often with Julia Schnebly, a solo act.

Most bookings are made by customers from Bellevue, and the least are in Auburn. As Auburn is a smaller city, the data is understandable.

Out of eight agents, half the agents have worked with 10 different entertainers, so it seems likely that the agents balance their attention among the entertainers.